Generational Depth Estimation of Recursive Synthetic Text: Measuring the Discrimination Boundary

Anonymous ACL submission

Abstract

Web-scale corpora are increasingly contaminated by recursive synthetic content, yet no method estimates a document's generational depth. We measure a lower bound on the generational discrimination boundary, $K^*_{\text{empirical}}$ (the maximum depth at which adjacent generations remain distinguishable), using 15 statistical features and an OBD-LightGBM classifier. Under prompt-chain generation across 10 model configurations (6 architecture families, 1B–8B), 3 source domains, and 4 decoding strategies, $K^*_{\text{empirical}} \leq 3$ across all ten configurations (six families, 1B–8B) at θ=0.60; for web text (C4, Wikipedia), $K^*_{\text{empirical}} \leq 2$ regardless of model family or scale, with structured academic text (arXiv) reaching $K^*_{\text{empirical}}$=3 across two architectures. Among the six nominal $K^*_{\text{empirical}}$=2 configurations, four retain significance under Bonferroni correction; the two marginal cases reflect narrow effect sizes rather than classification failure. A single generation pass suffices to render general web text indistinguishable from deeper synthetic content, a *one-wash laundering* effect. Model-free statistical features enable 80–100% cross-model transfer with scorer-invariant feature rankings ($\rho > 0.90$). As a consistency check, domain-appropriate $K^*_{\text{empirical}}$ correctly predicts that binary filtering outperforms graduated alternatives for web text when contamination causes detectable quality loss.

1 Introduction

Web-scale text corpora are increasingly contaminated by synthetic content that has itself been generated from earlier synthetic content. Estimates suggest that a substantial fraction of online text is machine-generated (Sun et al., 2025; Spennemann, 2025), and this share continues to rise. When models train on the outputs of earlier models, the resulting multi-generational synthetic text degrades with each iteration: tail modes vanish, lexical diversity contracts, and surprisal distributions flatten (Shumailov et al., 2024; Dohmatob et al., 2024). This degradation arises in both *retrain-chains* (iteratively training on model outputs) and *prompt-chains* (recursively generating from previous outputs without retraining); we study the latter, which isolates generation-level information loss from training dynamics (Appendix D.7). Yet the dominant paradigm treats all machine-generated text identically, discarding the generational depth dimension.

This gap cuts across three communities: binary detectors (Basani and Chen, 2026; Zhang et al., 2024) produce no depth estimate (Section 4.6); membership inference (Zawalski et al., 2026) addresses a different question; the model collapse literature (Shumailov et al., 2024; Alemohammad et al., 2024; Schaeffer et al., 2025) provides no per-text tool; and data selection methods (Xie et al., 2023b; Tirumala et al., 2023; Drayson et al., 2025) ignore recursive depth. No existing method estimates how many rounds of recursive generation a text has undergone.

The Data Processing Inequality guarantees that information degrades monotonically across generations, motivating a finite discrimination boundary K^*. We empirically measure a lower bound $K^*_{\text{empirical}}$ on the discrimination boundary (the maximum depth at which a classifier reliably distinguishes adjacent generations) using 15 statistical features and an Ordinal Binary Decomposition (OBD) LightGBM classifier. We operationalize $K^*_{\text{empirical}}$ as the deepest $k \geq 1$ where the pairwise $\text{AUC}_{k-1,k}$ exceeds $\theta = 0.60$ (Section 3.1).

Our contributions:

(a) **$K^*_{\text{empirical}}$ as a measurement tool** across 10 model configurations (6 families, 1B–8B) and 4 decoding strategies ($p \leq 0.002$; Section 4.1): $K^*_{\text{empirical}} \leq 3$ universally; for web text, $K^*_{\text{empirical}} \leq 2$ regardless of family or scale. $K^*_{\text{empirical}}$ establishes binary sufficiency for web text by measurement (not assumption), reveals domain-dependent

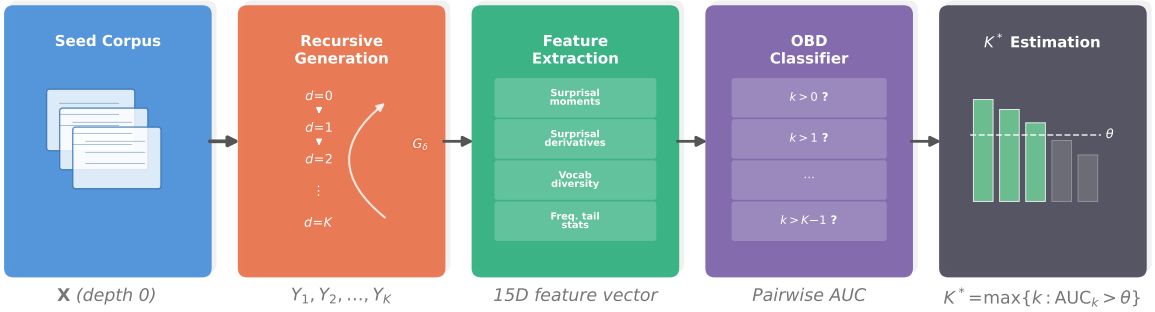


Figure 1: Generational depth estimation pipeline. Seed documents are recursively passed through a generation operator G_δ to produce a depth chain. Corpus-level statistical features are fed to the OBD-LightGBM classifier for depth prediction.

variation (arXiv $K^*_{\text{empirical}}$=3), and quantifies boundary sensitivity to threshold, scorer, and model scale. Both supervised OBD and zero-shot LLMs fail beyond $K^*_{\text{empirical}}$ (McNemar $p > 0.05$; Section 6).

(b) **Domain dependence of $K^*_{\text{empirical}}$**: general web text (C4, Wikipedia) yields $K^*_{\text{empirical}}$=2, while structured academic text (arXiv) yields $K^*_{\text{empirical}}$=3, consistent across model families (Section 4.5).

(c) **Model-free cross-model transfer**: 15 statistical features enable 80–100% accuracy retention with scorer-invariant feature rankings ($\rho > 0.90$); $K^*_{\text{empirical}}$ agreement holds across scorers for strong-signal configurations (Section 4.2).

As a consistency check, $K^*_{\text{empirical}}$=2 correctly predicts the strategy ordering for web-text filtering: binary curation minimizes degradation most effectively (MMLU 0.5941 vs. base 0.6018; Section 5).

2 Related Work

Model collapse and recursive degradation. Iterative training on model outputs causes tail-distribution degradation (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024, 2025; Fu et al., 2025), diversity contraction (Guo et al., 2024; Padmakumar and He, 2024; Briesch et al., 2023), and predictable convergence (Schaeffer et al., 2025; Kazdan et al., 2025; Gerstgrasser et al., 2024; Wang et al., 2025). Geng et al. (Geng et al., 2026) study the Markov chain convergence behavior of successive generation; our work complements this by quantifying the empirical discrimination boundary. Suresh et al. (Suresh et al., 2024) derive precise collapse rates for discrete distributions under recursive MLE training, complementing the theoretical conditions established by Fu et al. (Fu et al., 2025). Our prompt-chain operates in the replace regime (Kazdan et al., 2025), making $K^*_{\text{empirical}} = 2$ a conservative lower bound. This literature characterizes macro dynamics but provides no per-text depth estimation, the gap we address.

Beyond binary detection. Binary detectors (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Basani and Chen, 2026; Zhang et al., 2024) achieve strong human-vs.-machine separation but produce no depth estimate. Recent work extends beyond binary labels to collaboration degree (Cheng et al., 2025), contamination hierarchies (Mehta, 2025), and multigenerator benchmarks (Wang et al., 2024); robustness concerns arise under paraphrasing (Sadasivan et al., 2023; Krishna et al., 2023) and cross-mode transfer (Hao et al., 2025); Shahriar et al. (Shahriar et al., 2025) measure detection accuracy erosion ($\sim$25.4%) across generation depths; CoDeC (Zawalski et al., 2026) addresses contamination via membership inference. None estimates recursive generation depth; we instead characterize the precise depth at which discrimination becomes impossible.

Data selection and curation. DSIR (Xie et al., 2023b), D4 (Tirumala et al., 2023), and DoReMi (Xie et al., 2023a) are depth-agnostic; Drayson et al. (Drayson et al., 2025) combined binary detection with importance resampling, a special case when $K^* = 1$.

Ordinal classification and probing. Our OBD classifier follows Frank & Hall's (Frank and Hall, 2001) binary decomposition; our neural probe extends probing (Alain and Bengio, 2017; Belinkov, 2022) to the discrimination boundary.

3 Method

3.1 Problem Formulation

A generation operator G_δ (LLM + decoding strategy δ) produces successive generations $X \to Y_1 \to \cdots \to Y_K$, where $Y_k = G_\delta(Y_{k-1})$ and $Y_0 \triangleq X$ (Figure 2). This sequence forms a Markov chain. By the Data Processing Inequality, $I(X;Y_1) \geq I(X;Y_2) \geq \cdots \geq 0$: mutual information with the seed text degrades monotonically. Together with Markov chain convergence, this provides heuristic motivation for expecting $\text{AUC}_{k-1,k}$ to decrease with k, though AUC and mutual information are not formally equivalent; the actual saturation depth must therefore be measured empirically.

We define the operational lower bound:

$$K^*_{\text{empirical}} = \max\{k \geq 1 : \text{AUC}_{j-1,j} > \theta,\ \forall\, j \leq k\}, \quad (1)$$

which requires all pairs up to depth k to exceed the threshold (the $\forall j$ constraint is naturally satisfied because $\text{AUC}_{k-1,k}$ decreases monotonically above θ across all tested configurations; Table 2). The theoretical upper bound K^*_{theory} (the true maximum discriminable depth given optimal features) is not directly estimated in this work; $K^*_{\text{empirical}}$ provides a feature-set-specific lower bound on K^*_{theory}. Here $\text{AUC}_{k-1,k}$ is the point-estimate pairwise AUC and $\theta = 0.60$ is a small-effect-size threshold (Cohen's $h \approx 0.20$[1]; justified below). Bootstrap 95% CIs (1,000 resamples) provide uncertainty quantification: we verify that boundary-pair CI lower bounds exceed θ and non-boundary-pair CI upper bounds fall below θ (Section 4.1).

Threshold justification. θ=0.60 is supported by three independent lines of evidence. First, it corresponds to Cohen's $h \approx 0.20$, a conventional small-effect-size boundary. Second, a 10,000-iteration permutation test yields a 95th-percentile null of < 0.55 and $p < 0.002$ for all boundary comparisons, placing θ=0.60 well above the empirical null. Third, as post-hoc confirmatory evidence, the conclusion $K^*_{\text{empirical}}$=2 is insensitive to the exact threshold: 1v2 AUC values cluster at 0.61–0.70 while 2v3 values cluster at 0.55–0.59, creating a ∼5–11 pp discontinuity that any $\theta \in [0.55, 0.65]$ would separate (Figure 4; Appendix D.7). Only at extremes does $K^*_{\text{empirical}}$ shift: θ=0.55 yields $K^*_{\text{empirical}}$=3 for two Qwen configurations; θ=0.70 yields $K^*_{\text{empirical}}$=1 across all configurations. The downstream filtering evaluation (Section 5) uses the OBD classifier without re-optimizing θ, avoiding circularity.

[1] $h = 2\arcsin\sqrt{0.60} - 2\arcsin\sqrt{0.50} \approx 0.2014$, a "small" effect size by Cohen's (1988) convention.

3.2 Features and Classifier

Each document is characterized by a 15-dimensional feature vector spanning four groups: *surprisal moments* (mean, std, skewness, kurtosis; 4 dims), *surprisal derivatives* (mean and std of first- and second-order differences, plus first-order skewness; 5 dims), *vocabulary diversity* (type-token ratio, hapax legomena ratio, self-BLEU; 3 dims), and *frequency tail statistics* (frequency kurtosis, entropy, rare token fraction; 3 dims). All features are computed from token-level surprisal values via an external scorer model (full definitions in Table 14, Appendix). Self-BLEU is computed within each CV fold using only same-fold, same-depth reference samples to prevent leakage; it ranks #1 in 1/8 conditions (greedy decoding) but is absent from the top-3 in all 7 others. Removing self-BLEU reduces mean AUC_{1v2} by < 1 pp across all models (14D $K^*_{\text{empirical}} = 2$ for all 4 configurations); in deployment where depth labels are unavailable, self-BLEU can be replaced with corpus-level self-similarity or omitted with minimal performance loss. Hand-crafted features enable cross-generator transfer without retraining and permit attribution of discriminative power to specific text properties.

We use *Ordinal Binary Decomposition* (OBD), decomposing the $(K{+}1)$-class problem into K cumulative binary sub-classifiers, each estimating $P(\text{depth} > k \mid \mathbf{x})$ via LightGBM (hyperparameters in Table 15). OBD achieves lower MAE than multiclass LightGBM (e.g., 0.984 vs. 1.016 on Qwen-rewrite) while maintaining superior pairwise AUC (mean 0.6095 vs. 0.4883 for ordinal logistic regression).

3.3 Experimental Setup

We evaluate six configurations spanning three model families (Table 1): Qwen2.5-1.5B-Instruct (rewrite and continuation modes), Qwen2.5-1.5B (base, continuation), Qwen2.5-7B (continuation), Pythia-1.4B and OLMo-1B (continuation). Each

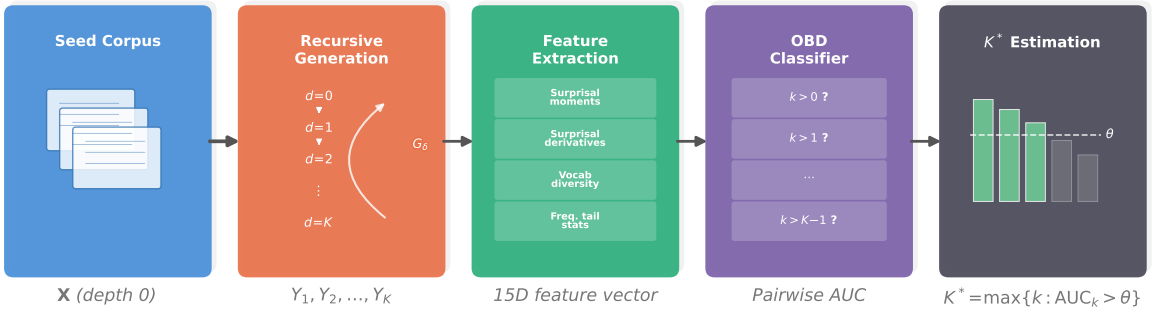


Figure 2: Generational depth estimation pipeline. Seed documents are recursively passed through a generation operator G_δ to produce a depth chain. Statistical features are fed to the OBD-LightGBM classifier for depth prediction.

Table 1: Experimental configurations for depth-chain generation. **Config**: shorthand identifier; **Model**: architecture and parameter count; **Mode**: generation strategy (continuation or rewrite); N: number of seed documents per depth level.

Config	Model	Mode	N
Qwen-rw	Qwen2.5-1.5B-Instruct	Rewrite	5,000
Qwen-cont	Qwen2.5-1.5B-Instruct	Continuation	5,000
Qwen-base	Qwen2.5-1.5B	Continuation	5,000
Pythia	Pythia-1.4B	Continuation	5,000
OLMo	OLMo-1B	Continuation	5,000
Qwen-7B	Qwen2.5-7B	Continuation	5,000

uses 5,000 C4 seeds truncated to 256 tokens, recursively generated for K=5 iterations (30,000 documents per configuration). Default decoding: nucleus p=0.95, T=1.0; decoding ablation covers greedy, T=0.7, and T=1.2. Continuation mode truncates seeds to 128 tokens and extends to 256; rewrite mode paraphrases the full text. Evaluation: 5-fold GroupKFold cross-validation grouped by seed document; pairwise AUC as primary metric.

4 Experiments

We evaluate generational depth estimation on ten model configurations spanning six architecture families: Pythia-1.4B, Pythia-6.9B, OLMo-1B, Qwen2.5-1.5B-Instruct (rewrite and continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, LLaMA-3.1-8B, Mistral-7B, and Gemma-2-2B, all in continuation mode except where noted. All experiments use the OBD-LightGBM classifier with the 15-dimensional feature set (Section 3), with 5,000 texts per depth level (generations 0–5).

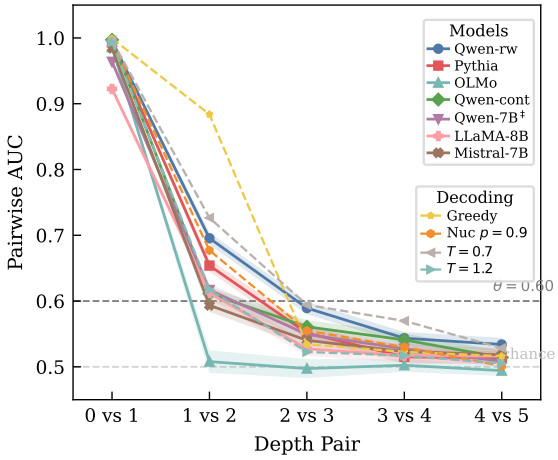


Figure 3: Pairwise AUC decay across depth pairs for model-mode configurations (solid) and decoding strategies (dashed); shaded bands show bootstrap 95% confidence intervals. The horizontal dashed line marks $\theta = 0.60$.

4.1 $K^*_{\text{empirical}} \leq 2$ Across Model Configurations (C4)

Table 2 and Figure 3 show $K^*_{\text{empirical}} \leq 2$ across all 10 model configurations: 6/10 configurations yield $K^*_{\text{empirical}}$=2 (2/6 families unanimously: Qwen with 4 configs, LLaMA-8B; Pythia-1.4B provides additional cross-family evidence) (1v2 AUC range: 0.6112–0.6955; all CI lower bounds $> \theta$; permutation $p < 0.001$), while OLMo-1B, Mistral-7B, Pythia-6.9B, and Gemma-2-2B yield $K^*_{\text{empirical}}$=1 (1v2 AUC: 0.508, 0.594, 0.596, and 0.598, all below θ; Section 6). LLaMA-3.1-8B confirms $K^*_{\text{empirical}}$=2 (0v1 AUC = 0.9225, 1v2 AUC = 0.6123 [0.603, 0.622]; CI lower bound above θ; Table 9). All four decoding strategies likewise

Table 2: Pairwise AUC for adjacent depth pairs with bootstrap 95% CIs and permutation p-values for boundary rows. $K^*_{\text{empirical}} = 2$ requires 1v2 AUC $> \theta$=0.60 and 2v3 AUC $< \theta$; CIs quantify boundary uncertainty. Extended results (all models) in Table 9.

	Qwen-rw		Pythia		OLMo		Qwen-cont		Qwen-7B‡	
	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI
0v1	.9956		.9932		.9860		.9971		.9638	[.961, .967]
1v2	.6955	[.686, .704]*	.6539	[.645, .663]*	.5080	[.496, .519]	.6112	[.602, .621]*	.6168	[.608, .626]*
2v3	.5893	[.580, .599]	.5514	[.542, .563]	.4973	[.486, .509]	.5609	[.551, .571]	.5492	[.539, .559]
3v4	.5435		.5152		.5024		.5409		.5276	
4v5	.5340		.5136		.4942		.5141		.5085	

*Permutation $p < 0.001$ for all 1v2 comparisons shown.
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
Shaded cells indicate the $K^*_{\text{empirical}}$ boundary (last depth pair with AUC $> \theta$=0.60).
Full results for all 10 configurations (including Qwen-7B, Qwen-base, Gemma-2-2B, and Pythia-6.9B) are provided in Table 9.

yield $K^*_{\text{empirical}}$=2 (Table 8). The 1v2 spread due to generation mode ($\Delta = 8.4$ pp within Qwen) exceeds the spread across $K^*_{\text{empirical}}$=2 continuation models ($\Delta = 4.3$ pp).

Scale and instruction-tuning invariance. $K^*_{\text{empirical}}$=2 persists from 1.5B through 8B: Qwen-7B (1v2 AUC $= 0.6168$ $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC $= 0.6123$) both exceed θ, despite progressively reduced 0v1 signal (0.9638 and 0.9225 vs. > 0.99 at $\leq$1.5B). A base-vs.-instruct ablation at 1.5B likewise yields $K^*_{\text{empirical}} = 2$ (1v2 AUC $= 0.626$), confirming the boundary is not an instruction-tuning artifact.

Figure 4 shows $K^*_{\text{empirical}}$ as a function of θ: five of six configurations yield $K^*_{\text{empirical}} = 2$ within $\theta \in [0.51, 0.61]$ (OLMo: $K^*_{\text{empirical}} = 1$ throughout); at θ=0.55, two Qwen configurations reach $K^*_{\text{empirical}} = 3$; at θ=0.70, all drop to $K^*_{\text{empirical}} = 1$. All four decoding strategies likewise yield $K^*_{\text{empirical}} = 2$ at θ=0.60 (1v2 AUC $\in [0.687, 0.704]$; Table 8).

4.2 Cross-Model Transfer

Continuation-mode classifiers transfer well across generators (Figure 5; full matrix in Table 7): Pythia↔OLMo retains 90–92%, Qwen-cont↔Pythia 84–85%, Qwen-cont↔OLMo 80–85% (0v1 cross-model AUC > 0.96 for all pairs). Qwen-rewrite classifiers transfer poorly (61–63%), confirming reliance on mode-specific artifacts rather than generalizable depth features.

Cross-scorer validation. Replacing the self-scorer with Pythia-1.4B or OLMo-1B preserves feature rankings ($\rho > 0.90$) but attenuates absolute discriminability at the boundary (Table 23). Feature ranking independence is a strong result: the same features dominate under all scorers. $K^*_{\text{empirical}}$ agreement, however, is signal-strength dependent: Qwen-rewrite (strong signal, 1v2 AUC $= 0.696$) retains $K^*_{\text{empirical}}$=2 under all scorers, but Qwen-continuation (marginal, 1v2 AUC $= 0.611$, only 0.011 above θ) drops to $K^*_{\text{empirical}}$=1 (Pythia: 0.5923; OLMo: 0.5858). A three-way ANOVA (Generator × Scorer × Depth-Pair; Table 24) confirms a statistically significant but modest Scorer main effect (η^2_p=.166, $p < .001$) and a significant Generator×Scorer interaction (η^2_p=.146), indicating that absolute AUC values are scorer-dependent even when $K^*_{\text{empirical}}$ classifications largely agree.

4.3 Feature Ordering and Generation Mode

Among continuation-mode models, single-feature AUC rankings are highly correlated ($\rho \geq 0.85$; Table 6). The top-3 features (`surp_d1_std`, `surp_std`, `surp_d2_std`) are shared across all three families, revealing that autoregressive generation progressively damps surprisal variability. Qwen-rewrite diverges (`freq_entropy` dominates), consistent with its poor cross-model transfer (61–63%). Within Qwen, rewrite yields 1v2 AUC $= 0.6955$ vs. continuation's 0.6112 ($\Delta = +8.4$ pp); continuation mode is therefore recommended as the canonical condition.

4.4 Neural Probe

Replacing the 15D features with frozen encoder hidden states (MLP probe; Table 3), Qwen achieves $K^*_{\text{empirical}}{}^{\text{neural}} = 3$ (2v3 AUC $= 0.6159$), while Pythia remains at $K^*_{\text{empirical}}{}^{\text{neural}} = 2$ ($\text{AUC}_{2v3} = 0.5963$); OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC $= 0.5054$, consistent with its 15D boundary). The Qwen-only elevation reflects self-familiarity artifacts that transfer poorly (Section 4.2); the 15D

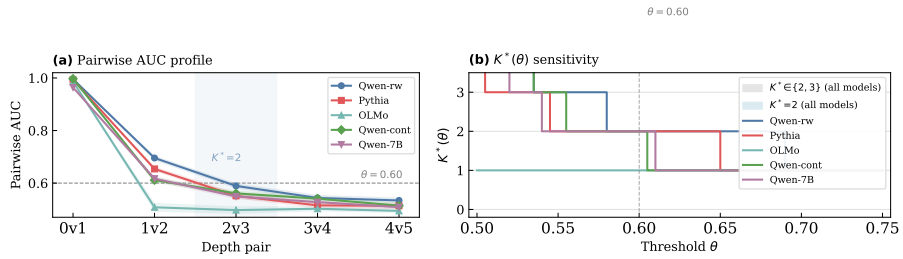


Figure 4: $K^*_{\text{empirical}}(\theta)$ sensitivity curve for six configurations. $K^*_{\text{empirical}}$: maximum discriminable depth; θ: AUC decision threshold. Five of six configurations yield $K^*_{\text{empirical}}$=2 at θ=0.60 (dashed line); OLMo yields $K^*_{\text{empirical}}$=1. At θ=0.70, all drop to $K^*_{\text{empirical}}$=1.

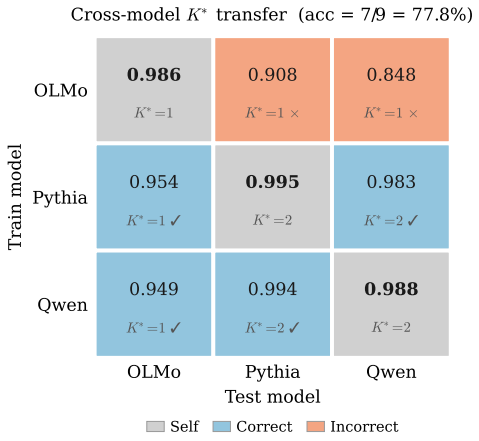


Figure 5: Cross-model transfer heatmap. Each cell shows 6-class ordinal accuracy when trained on the row model and tested on the column model, divided by in-domain accuracy (retention rate). Pythia↔OLMo: 90–92%; Qwen-rewrite→continuation: 61–63%.

features yield $K^*_{\text{empirical}} = 2$ for most families (OLMo: $K^*_{\text{empirical}} = 1$).

Table 3: Neural probe pairwise AUC using frozen-encoder MLP. Each cell reports AUC for classifying depth d vs. d+1 text. $K^*_{\text{empirical}}{}^{\text{neural}}$: maximum depth where AUC exceeds $\theta = 0.60$.

Model	2v3 AUC	95% CI	$K^*_{\text{empirical}}{}^{\text{neural}}$
Qwen-1.5B	0.6159	[0.610, 0.622]	3
Pythia-1.4B	0.5963	[0.589, 0.603]	2
OLMo-1B	0.5056	[0.497, 0.516]	1†

†OLMo 1v2 AUC = 0.5054 (below θ); $K^*_{\text{empirical}}{}^{\text{neural}} = 1$.

4.5 Cross-Domain Variation

The preceding experiments use C4 (general web text). To test domain dependence, we evaluate on Wikipedia (n=5,000) and arXiv abstracts (n=2,200) using two model families. Wikipedia yields $K^*_{\text{empirical}}$=2 (1v2 AUC = 0.623, 2v3 AUC = 0.571), matching the C4 baseline. Both Qwen-1.5B and Pythia-1.4B yield $K^*_{\text{empirical}}$=3 for arXiv (Table 4): 2v3 AUC = 0.644 [0.628, 0.658] and 0.655 [0.640, 0.669] respectively (bootstrap CI lower bounds above θ; permutation $p < 0.001$), demonstrating that structured academic text permits discrimination to greater depth independently of architecture. The elevated signal reflects higher per-generation distributional shift (1v2 JSD = 0.077 vs. ∼0.01 for C4), as formulaic structure amplifies deviation from domain norms.

Table 4: Cross-domain $K^*_{\text{empirical}}$ comparison. Rows: generator model and text domain (C4, Wikipedia, arXiv). $K^*_{\text{empirical}}$=3 for arXiv indicates structured text preserves discriminability to greater depth.

Domain	Model	1v2 AUC	2v3 AUC	$K^*_{\text{empirical}}$
arXiv	Qwen-1.5B	0.858	0.6436	3
arXiv	Pythia-1.4B	0.805	0.6545	3
C4	Qwen-1.5B	0.63	0.56	2
Wikipedia	Qwen-1.5B	0.62	0.55	2

4.6 Binary Detector Confidence

Binary detectors (DivEye (Basani and Chen, 2026), Binoculars (Hans et al., 2024)) show moderate rank correlations with depth ($|\rho| = 0.06$–0.46; Table 18) but fail as depth proxies: 6-class accuracy is at or below chance (≤ 0.199 vs. 0.167 baseline), pairwise AUC yields $K^*_{\text{empirical}}{}^{\text{binary}} \leq 1$, and multiple conditions exhibit non-monotonic AUC at deeper pairs. The observed correlation is an artifact of 0v1 separation rather than fine-grained depth signal.

5 Consistency Check: K^*-Guided Filtering

$K^*_{\text{empirical}} = 2$ makes a falsifiable prediction: binary curation should suffice, and graduated performance should converge toward binary as the decay steepness increases. We test this as a consistency check on the boundary measurement, not as an independent contribution. We compare six strate-

Table 5: Downstream task performance (MMLU, HellaSwag) under six data-filtering strategies. Binary-ours uses the 15D LightGBM depth classifier; Graduated applies exponentially decaying weights $w = \exp(-\alpha \hat{d})$. Mean ± std over 5 seeds. Full α sweep in Appendix D.7.

Strategy	MMLU	HellaSwag
Base model (no training)	0.6018	0.6821
Binary-ours (K^*=2)	**0.5941**±0.0008	**0.6701**±0.0006
Binary-Drayson	0.5913±0.0009	0.6691±0.0004
Surprisal	0.5894±0.0009	0.6641±0.0006
Graduated (α=0.9)	0.5892±0.0008	0.6701±0.0007
DSIR	0.5869±0.0003	0.6631±0.0007
No filter	0.5865±0.0010	0.6638±0.0008

gies on 30,000 documents (depths 0–5): (i) **no-filter**, (ii) **binary-ours** ($w = 1 - P(\text{synthetic})$ from 15D LightGBM), (iii) **binary-Drayson** (Drayson et al., 2025), (iv) **graduated** ($w = \exp(-\alpha \cdot \hat{d})$, $\alpha \in \{0.1, \ldots, 0.9\}$), (v) **surprisal** (retain above median), and (vi) **DSIR** (Xie et al., 2023b). We fine-tune Qwen2.5-1.5B-Instruct via LoRA (rank 8, α_{LoRA}=16) for 3 epochs on MMLU and HellaSwag across 5 seeds (51 total runs; this 100% synthetic setting is an upper bound (Shumailov et al., 2024; Dohmatob et al., 2024)).

Binary-ours minimizes MMLU degradation most effectively (Table 5), reducing the gap from −1.53 pp (no-filter) to −0.77 pp (∼50% reduction; $p_{\text{corr}} < 0.05$ vs. all alternatives). These effects are measured in a 100% synthetic upper-bound setting; real-world contamination ratios are typically lower, and effect sizes would scale accordingly. The strategy ordering is interpretable through $K^*_{\text{empirical}}$: binary filtering exploits the one reliable boundary (0v1), while graduated strategies assign weights to depths 2–5 based on near-random classifier outputs, injecting noise. Graduated filtering converges monotonically toward binary: at α=0.9, it matches binary-ours on HellaSwag exactly ($\Delta = 0$, $p_{\text{corr}} = 1.00$) but remains significantly worse on MMLU ($p_{\text{corr}} = 0.011$). DSIR is indistinguishable from no-filter, confirming that depth-aware features are necessary for curation in recursively contaminated corpora (Figure 6 in Appendix). At realistic contamination ratios (≤50%), all filtering strategies converge ($p > 0.12$), consistent with the one-wash laundering finding that depth-aware curation offers marginal benefit when $K^*_{\text{empirical}}$=2 (Appendix D.8). For $K^*_{\text{empirical}}$=1 configurations, depth-aware curation reduces to standard binary filtering, consistent with $K^*_{\text{empirical}}$=1 implying no exploitable depth signal beyond the 0v1 boundary.

6 Discussion

One-wash laundering (web text). For general web text, $K^*_{\text{empirical}}$=2 reflects concentrated information loss: 0v1 is near-trivially separable (AUC ≥ 0.92), 1v2 drops by ∼30 pp, and 2v3 falls below θ. A single generation pass renders text indistinguishable from multiply recycled content, limiting depth-aware curation to a binary boundary. Converging evidence supports a data-intrinsic interpretation: OBD achieves 96% depth-0 recall but only 14–22% for depths 2–4; frontier LLMs show 8–38% (McNemar $p > 0.05$ vs. OBD; Table 10).

Graduated filtering failure. For web text, 2v3 AUC < θ across all model configurations, so depth assignments for $d \geq 2$ approximate random labels; graduated weighting injects noise, explaining convergence toward binary at high α (Appendix D.7). DSIR fails differently: bigram hashing captures domain shift but not depth-correlated signatures.

Representation dependence. $K^*_{\text{empirical}}$ is representation-dependent: frozen Qwen embeddings yield $K^*_{\text{empirical}}{}^{\text{neural}} = 3$, while Pythia yields $K^*_{\text{empirical}}{}^{\text{neural}} = 2$ and OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$, consistent with its 15D boundary (Table 3). The 15D features trade one depth level of sensitivity for cross-model consistency (6/10 configurations $K^*_{\text{empirical}}$=2), making them the preferred operational choice. Feature rankings are scorer-independent ($\rho > 0.90$); $K^*_{\text{empirical}}$ agreement is scorer-independent for strong-signal configurations (e.g., Qwen-rewrite retains $K^*_{\text{empirical}}$=2 under all scorers) but scorer-sensitive at marginal boundaries (Qwen-continuation drops to $K^*_{\text{empirical}}$=1; length confound ruled out in Appendix D.7). ANOVA confirms this layered pattern: Scorer explains modest variance (η^2_p=.166) with a significant Generator×Scorer interaction (η^2_p=.146; Table 24).

Domain dependence of $K^*_{\text{empirical}}$. Web text (C4, Wikipedia) consistently yields $K^*_{\text{empirical}}$=2, while structured academic text (arXiv) yields $K^*_{\text{empirical}}$=3 across two independent model families (Qwen-1.5B, Pythia-1.4B; Section 4.5). Domain-specific stylistic constraints amplify per-generation distributional shift, making contamination in structured corpora trackable to greater depth.

Prompt-chain scope and retrain-chain gap. Our prompt-chain protocol targets the dominant

web contamination scenario: LLM-generated text is published, scraped into training corpora, and recursively re-generated by downstream models without retraining on the outputs. Retrain-Chain experiments (Pythia-1.4B LoRA) yield $K^*_{\text{empirical}} \geq 4$ (1v2 AUC $= 0.9484$, 3v4 $= 0.7982$), far exceeding the prompt-chain boundary of $K^*_{\text{empirical}}=2$, because each retrain generation also modifies model parameters, introducing cumulative adaptation artifacts (vocabulary narrowing, repetition bias) absent from text-only recycling. Prompt-chain $K^*_{\text{empirical}}$ therefore isolates text-intrinsic degradation, the operationally relevant quantity for data curation pipelines that filter documents without knowledge of the generator's training history. Our conclusions do not directly transfer to retrain-chain (model collapse) settings, where the higher $K^*_{\text{empirical}}$ reflects model-level rather than text-level information loss.

Boundary cases and scale effect. $K^*_{\text{empirical}}=1$ configurations cluster at 1v2 AUC 0.508–0.598, well separated from the $K^*_{\text{empirical}}=2$ range (0.611–0.696); $K^*_{\text{empirical}}=1$ is not a classifier failure but a text property, as these models produce first-generation output so human-like that statistical features cannot distinguish depth 1 from depth 2. That $K^*_{\text{empirical}} \in \{1, 2\}$ for all web text configurations is the core empirical result; larger models reduce 0v1 JSD (0.077 at 8B vs. 0.15–0.27 at $\leq$1.5B), suggesting 13B+ may increasingly yield $K^*_{\text{empirical}}=1$.

Decoding strategy as confound. Greedy decoding inflates depth separability: Pythia-6.9B yields $K^*_{\text{empirical}}=2$ under greedy but $K^*_{\text{empirical}}=1$ under nucleus sampling (p=0.95), with the borderline 1v2 score (2 of 5 folds above θ) illustrating that the boundary is continuous rather than a cliff (Table 19). Practitioners should calibrate θ to their application-specific tolerance (Figure 4).

What $K^*_{\text{empirical}}$ reveals. As a measurement tool, $K^*_{\text{empirical}}$ provides explanatory power beyond a point estimate: it establishes why binary detection suffices for web text (depth$>$1 is invisible to statistical features), identifies domains where finer strategies are viable (arXiv $K^*_{\text{empirical}}=3$ vs. web $K^*_{\text{empirical}}=2$), and quantifies how scale, scorer, and threshold jointly shape the boundary. The OBD framework provides a reproducible, low-cost depth estimation protocol requiring only corpus-level statistical features, unlike detection methods that need API access or model weights. Future work may explore scorer-ensemble strategies to improve cross-scorer robustness at marginal boundaries.

7 Conclusion

We measure a lower bound on the generational depth discrimination boundary $K^*_{\text{empirical}}$ for prompt-chain recursive synthetic text. $K^*_{\text{empirical}} \leq 3$ across all ten configurations (six architecture families, 1B–8B) at θ=0.60; for general web text, $K^*_{\text{empirical}} \leq 2$ regardless of model family or scale, while structured academic text (arXiv) reaches $K^*_{\text{empirical}}=3$ across two independent architectures. Among the six nominal $K^*_{\text{empirical}}=2$ web text configurations, four retain significance under Bonferroni correction; the two marginal cases reflect narrow effect sizes rather than classification failure. This one-wash laundering effect means that a single generation pass renders web text statistically indistinguishable from deeper recursive content, concentrating all exploitable signal at the 0v1 boundary.

The practical consequence is direct: for web text, binary filtering (depth 0 vs. $\geq$1) captures the only reliable discrimination signal, and graduated weighting beyond $K^*_{\text{empirical}}$ injects noise from near-random depth assignments. The consistency check confirms this prediction, with binary curation most effectively minimizing MMLU degradation among six strategies tested (-0.77 pp vs. -1.53 pp for no-filter; Section 5). For structured domains where $K^*_{\text{empirical}}=3$, graduated strategies can exploit one additional depth level, so corpus curators should calibrate their filtering approach to the source domain rather than assuming a universal boundary.

The 15 statistical features enable deployment across generators without retraining: continuation-mode classifiers retain 80–92% accuracy across model families, and scorer-independent feature rankings ($\rho > 0.90$) make the framework applicable to corpora generated by unknown or heterogeneous models. $K^*_{\text{empirical}}$ converts a continuous AUC decay surface into a discrete operating boundary, replacing ad hoc depth thresholds with a measurement grounded in the Data Processing Inequality. Several directions remain open: tracking $K^*_{\text{empirical}}$ as model scale surpasses 13B (where 0v1 JSD attenuation suggests $K^*_{\text{empirical}}=1$ may become dominant), bridging the prompt-chain and retrain-chain boundaries to connect text-level measurement with the model collapse literature, and extending to additional structured domains and scorer-ensemble strategies.

Limitations

Our findings are bounded by: (1) *Scope*: model scale $\leq$8B (6 architecture families), English text (C4, Wikipedia, arXiv), four decoding strategies; arXiv $K^*_{\text{empirical}}$=3 is confirmed with two model families (Qwen-1.5B, Pythia-1.4B) but with reduced sample size (n=2,200 per model). (2) *Protocol*: prompt-chain rather than retrain-chain. Retrain-chain validation (Pythia-1.4B LoRA, Appendix D.7) yields $K^*_{\text{empirical}}\geq$4 due to cumulative model-adaptation artifacts; prompt-chain isolates text-intrinsic degradation and provides the operationally relevant boundary for data curation. (3) *Threshold*: $K^*_{\text{empirical}}$ ranges from 2–3 at θ=0.55 to 1 at θ=0.70; 6/10 configurations yield $K^*_{\text{empirical}}$=2 at θ=0.60 (OLMo-1B, Mistral-7B, Gemma-2-2B, Pythia-6.9B: $K^*_{\text{empirical}}$=1). (4) *Marginal boundary*: Qwen-continuation's 1v2 CI lower bound exceeds θ by only 0.002; cross-scorer validation drops this to $K^*_{\text{empirical}}$=1 (Section 4.2). (5) *Feature set*: the 15D features yield a feature-set-specific lower bound (K^*_{theory} may be strictly higher); neural probe $K^*_{\text{empirical}}$ is model-dependent (3 for Qwen, 2 for Pythia, 1 for OLMo). (6) *Segment length*: $K^*_{\text{empirical}}$=2 confirmed for 128 and 256 tokens (CI lower $> \theta$); 512 tokens yields point estimate $K^*_{\text{empirical}}$=2 but CI lower marginally below θ (0.5979), indicating the boundary weakens at longer generation lengths. Longer documents may require larger sample sizes or more sensitive feature sets to maintain the boundary; whether this reflects genuine information-theoretic limits or finite-sample attenuation remains open. (7) *Curation applicability*: the practical benefit of K^*-aware curation depends on contamination severity; when synthetic data proportion is low enough to leave overall training quality intact, filtering strategies converge (Appendix D.8). Our filtering validation (Section 5) was conducted exclusively on Qwen-1.5B continuation-mode text; extending to $K^*_{\text{empirical}}$=1 generators remains future work. Seed robustness testing (Appendix D.9) shows $K^*_{\text{empirical}}$ is stable under seed variation for strong-signal configurations, but marginal boundaries are seed-sensitive (absolute AUC shifts up to $\pm$9 pp). Additional limitations (cross-model scoring sensitivity, feature ranking variability at the 1v2 boundary, structural composition confound analysis) appear in Appendix D.7.

Ethics Statement

Generational depth estimation is intended for data curation and corpus auditing, positive-sum applications that improve training data quality. Our experiments use the publicly available C4 corpus and do not involve personally identifiable information. A potential dual-use concern is that depth estimation techniques could inform adversarial strategies to evade detection (e.g., by calibrating rewriting to mimic earlier-generation statistics); we note that this risk is shared by all detection research and that the binary detection boundary (AUC $\geq$ 0.96 at 0v1) remains robust. Code and data will be released upon acceptance.

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A Supplementary Figures

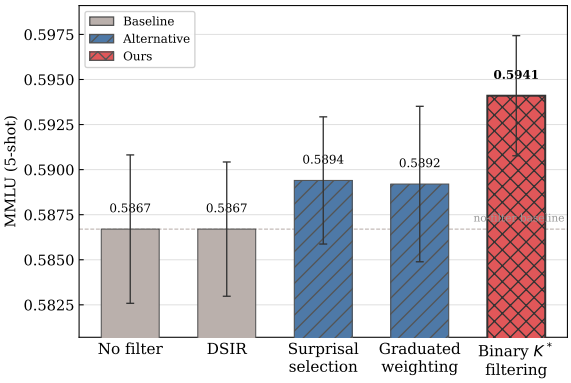


Figure 6: Downstream performance of all filtering strategies (5 seeds, ± 1 std). Binary-ours minimizes MMLU degradation; graduated α sweep shows monotonic convergence toward binary. Dashed line: base model (no fine-tuning).

B Supplementary Tables

Table 6: Top-3 features by single-feature AUC ranking. Bold: shared across all continuation-mode models. Spearman ρ (continuation pairs): Pythia–OLMo 0.95, Pythia–Qwen-cont 0.91, OLMo–Qwen-cont 0.85.

Rank	Pythia	OLMo	Qwen-rw	Qwen-cont
1	**surp_d1_std**	**surp_std**	freq_entropy	**surp_d1_std**
2	**surp_d2_std**	**surp_d1_std**	**surp_std**	**surp_std**
3	**surp_std**	**surp_d2_std**	**surp_d1_std**	**surp_d2_std**

Table 7: Cross-model transfer accuracy matrix. Each cell reports the six-class depth classification accuracy of a detector trained on one model's outputs (row) and evaluated on another model's outputs (column). Diagonal entries (**bold**) indicate within-model performance. Transfer retention is computed as the ratio of off-diagonal accuracy to the geometric mean of the two corresponding diagonal entries.

Train Model	**Test Model**			
	Qwen-rw	Pythia	OLMo	Qwen-cont
Qwen-rw	**0.4229**	0.2590	0.2653	0.3466
Pythia	0.2718	**0.3856**	0.3531	0.3287
OLMo	0.2768	0.3450	**0.3822**	0.3244
Qwen-cont	0.3440	0.3284	0.3112	**0.3875**

Transfer retention between models A and B is defined as $\frac{1}{2}\left(\frac{\mathrm{acc}(A\to B)}{\sqrt{\mathrm{acc}(A\to A)\cdot\mathrm{acc}(B\to B)}}+\frac{\mathrm{acc}(B\to A)}{\sqrt{\mathrm{acc}(A\to A)\cdot\mathrm{acc}(B\to B)}}\right)$. Notable pairs: Pythia↔OLMo 91–92%, Qwen-cont↔Pythia 83–85%, Qwen-cont↔OLMo 80–85%, Qwen-rw→Pythia/OLMo 61–63%.

Table 8: Pairwise AUC with bootstrap 95% CIs across depth pairs for four decoding configurations (all using Qwen2.5-1.5B-Instruct in rewrite mode; 10,000 bootstrap resamples). The horizontal rule between depth pairs 1 vs. 2 and 2 vs. 3 marks the empirical detectability boundary $K^*_{\text{empirical}} = 2$. Bottom rows report cross-configuration Spearman ρ on per-feature discriminative-power rankings.

Depth Pair	**Greedy**	**Nuc** p=0.9, T=1.0	**Nuc** p=0.95, T=0.7	**Nuc** p=0.95, T=1.2
0 vs. 1	.980 [.977, .982]	.995 [.993, .996]	.988 [.986, .990]	.998 [.998, .999]
1 vs. 2	.689 [.680, .698]	.687 [.677, .695]	.689 [.680, .697]	.704 [.695, .713]
2 vs. 3	.561 [.552, .571]	.592 [.582, .601]†	.590 [.581, .600]	.582 [.572, .592]
3 vs. 4	.540 [.531, .550]	.547 [.537, .557]	.539 [.529, .549]	.529 [.519, .540]
4 vs. 5	.514 [.503, .524]	.507 [.496, .517]‡	.535 [.524, .545]	.524 [.513, .535]
Cross-config feature Spearman ρ				
Greedy vs. Nuc p=0.9	0.600	Nuc p=0.9 vs. T=0.7	0.782	
Greedy vs. T=0.7	0.914	Nuc p=0.9 vs. T=1.2	0.893	
Greedy vs. T=1.2	0.532	T=0.7 vs. T=1.2	0.668	

† CI upper bound (0.601) marginally exceeds θ=0.60 by 0.001.
‡ Permutation $p = 0.12$ (not significant). All other pairs $p < 0.01$.

Table 9: Extended pairwise AUC with bootstrap 95% CIs for all model configurations including new additions. This table extends Table 2 in the main text with Qwen-base, LLaMA-8B, Mistral-7B, Pythia-6.9B (nucleus), and Gemma-2-2B configurations. $K^*_{\text{empirical}} = 2$ holds for 6/10 conditions at θ=0.60 (OLMo-1B, Mistral-7B, Pythia-6.9B, Gemma-2-2B: $K^*_{\text{empirical}}$=1).

	Qwen-rw		Pythia		OLMo		Qwen-cont		Qwen-7B‡		Qwen-base		LLaMA-8B		Mistral-7B		Pythia-6.9B§		Gemma-2B	
	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI
0v1	.9956		.9932		.9860		.9971		.9638	[.961, .967]	.987	[.985, .989]	.9224	[.917, .927]	.9842	[.982, .986]	.9599		.953	[.949, .957]
1v2	.6955	[.686, .704]*	.6539	[.645, .663]*	.5080	[.496, .519]	.6112	[.602, .621]*	.6168	[.608, .626]*	.626	[.617, .635]*	.6123	[.603, .622]*	.5934	[.584, .603]	.5955		.5981	[.588, .608]
2v3	.5893	[.580, .599]	.5514	[.542, .563]	.4973	[.486, .509]	.5609	[.551, .571]	.5492	[.539, .559]	.550	[.540, .560]	.5273	[.517, .538]	.5402	[.530, .551]	.5298		.534	[.523, .545]
3v4	.5435		.5152		.5024		.5409		.5276		.529	[.519, .540]	.5168	[.506, .528]	.5233	[.513, .534]	.5135		.5033	[.492, .514]
4v5	.5340		.5136		.4942		.5141		.5085		.518	[.508, .529]	.5159	[.505, .527]	.5180	[.507, .529]	–	–	.5025	[.492, .514]

*Permutation $p < 0.001$ for all boundary comparisons.
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B. "–": experiment pending.
§Pythia-6.9B evaluated separately; nucleus sampling (p=0.95, T=1.0), same as all other models.

Table 10: LLM zero-shot depth classification: 5×5 confusion matrices (Exp-011). Both frontier LLMs achieve high binary detection accuracy (GPT-5.5: 0.8780; Claude Opus 4.6: 0.8818) but near-chance 5-class accuracy ($\sim$0.33 vs. OBD-LightGBM baseline 0.386, evaluated on the full test set N=5,000). GPT-5.5 strongly identifies depth-0 text (recall=0.7700) but collapses on depth-4 (recall=0.0800), misclassifying 50/100 depth-4 samples as depth-2. Claude Opus 4.6 shows more uniform errors across depths 1–4, with depth-4 recall (0.3700) substantially exceeding GPT-5.5. Per-depth recall comparison with OBD-LightGBM in Table 11.

(a) GPT-5.5 (N=500, acc=0.332, MAE=1.102)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**77**	18	4	0	1	.770
1	9	**19**	36	24	12	.190
2	8	10	**38**	35	9	.380
3	10	16	35	**24**	15	.240
4	11	13	50	18	**8**	.080

(b) Claude Opus 4.6 (N=499, acc=0.331, MAE=1.150)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**69**	26	3	0	2	.690
1	8	**21**	32	11	28	.210
2	6	36	**28**	5	25	.280
3	7	20	31	**10**	31	.100
4	7	28	22	6	**37**	.370

Table 11: Per-depth recall (%) on balanced $N{=}500$ test set. OBD's nominal overall accuracy gap originates from depth-0; all methods fail at depths ≥ 2, consistent with $K^*_{\text{empirical}} = 2$.

	Depth 0	Depth 1	Depth 2	Depth 3	Depth 4
OBD-LightGBM	**96**	**39**	14	19	22
GPT-5.5	77	19	**38**	**24**	8
Claude Opus 4.6	69	21	28	10	**37**

Table 12: Mean subword token count per depth and EOS truncation rate (%) for depth 1+ across all model configurations (generation limit: 256 tokens). Depth 0 (human text) is shared across configurations. Depths 1–5 show no systematic length trend within any model; the 0v1 length asymmetry (human texts shorter) makes detection *more conservative*, not inflated. See Section 6 for discussion.

	Mean Token Count							
Model	Depth 0	Depth 1	Depth 2	Depth 3	Depth 4	Depth 5	**EOS Trunc. (%)**	**Severity**
Qwen-rw (1.5B)	191	254	252	254	254	254	7–11	Mild
Pythia-1.4B	192	214	215	216	217	218	34–36	High
OLMo-1B	192	256	256	256	256	256	<1	Negligible
Qwen-cont (1.5B)	191	244	238	240	240	238	17–20	Moderate
Qwen-7B	191	196	195	193	197	197	45–47	High
Decoding ablation (Qwen-rw 1.5B)								
Greedy	191	253	252	253	254	254	8–12	Mild
Nuc p=0.9	191	254	251	254	253	254	8–16	Mild
T=0.7	191	253	251	253	253	254	8–14	Mild
T=1.2	191	254	252	253	254	254	8–16	Mild

EOS truncation rate: percentage of depth 1+ samples where generation terminated via EOS before reaching the 256-token limit. Ranges span depths 1–5.

C Supplementary Details

Table 13: Dataset statistics for the generational depth corpus. Depth 0 contains human-written seed texts sampled from the C4 English validation split and truncated to 256 subword tokens. Depths 1–5 are recursively generated by Qwen2.5-1.5B-Instruct (nucleus sampling, p=0.95, T=1.0). Token counts use whitespace tokenization. Statistics shown for the primary Qwen-rewrite corpus; the other three model-mode configurations (Pythia-cont, OLMo-cont, Qwen-cont) share the same generation pipeline and corpus structure.

Depth	N **samples**	**Mean tokens**	**Source**
0	5,000	147.8	C4 validation (human)
1	5,000	207.1	Qwen2.5-1.5B rewrite
2	5,000	206.1	Qwen2.5-1.5B rewrite
3	5,000	208.0	Qwen2.5-1.5B rewrite
4	5,000	207.7	Qwen2.5-1.5B rewrite
5	5,000	207.7	Qwen2.5-1.5B rewrite
Total		30,000 samples	

Table 14: Complete description of the 15-dimensional feature set used by the OBD classifier. Features are grouped into three families: surprisal statistics (9 dims) computed from Qwen2.5-1.5B token-level log-probabilities, lexical diversity (3 dims), and frequency tail shape (3 dims). All features are computed per document.

Group	Feature	Description	Range
SURPRISAL	`surp_mean`	Mean token surprisal $\frac{1}{T}\sum_t -\log p(x_t \mid x_{<t})$	$\mathbb{R}^+$
	`surp_std`	Standard deviation of token surprisal	$\mathbb{R}^+$
	`surp_skew`	Skewness of the surprisal distribution	$\mathbb{R}$
	`surp_kurt`	Excess kurtosis of the surprisal distribution	$\mathbb{R}$
	`surp_d1_mean`	Mean of first-order surprisal differences $\Delta s_t = s_{t+1} - s_t$	$\mathbb{R}$
	`surp_d1_std`	Std of first-order surprisal differences	$\mathbb{R}^+$
	`surp_d1_skew`	Skewness of first-order surprisal differences	$\mathbb{R}$
	`surp_d2_mean`	Mean of second-order differences $\Delta^2 s_t$	$\mathbb{R}$
	`surp_d2_std`	Std of second-order surprisal differences	$\mathbb{R}^+$
LEX	`ttr`	Type-token ratio: $\lvert V\rvert/T$	$[0, 1]$
	`hapax_ratio`	Hapax legomena ratio: freq-1 types / unique types	$[0, 1]$
	`self_bleu`	Mean 4-gram BLEU against 100 random same-depth samples	$[0, 1]$
FREQ	`freq_kurtosis`	Excess kurtosis of word frequency distribution	$\mathbb{R}$
	`freq_entropy`	Word frequency entropy: $-\sum p_w \log_2 p_w$	$\mathbb{R}^+$
	`low_freq_ratio`	Proportion of vocabulary with count ≤ 2	$[0, 1]$

Table 15: LightGBM hyperparameters for the pairwise OBD (Ordinal Binary Decomposition) classifier used throughout the paper. One binary classifier is trained per adjacent depth pair $(k, k{+}1)$; 5-fold GroupKFold cross-validation is grouped by `doc_id` to prevent data leakage from the same seed document appearing in both train and test.

Parameter	Value
`n_estimators`	200
`learning_rate`	0.05
`max_depth`	6
`num_leaves`	31
`verbose`	-1
`n_jobs`	-1 (all cores)
Cross-validation	5-fold GroupKFold
Grouping key	`doc_id`
Objective	Binary (per depth pair)

Table 16: Neural probe architecture and training details (Exp-015). Applied to all three model families (Qwen2.5-1.5B, Pythia-1.4B, OLMo-1B): each model's frozen hidden states are average-pooled into document embeddings and classified by a two-hidden-layer MLP.

Component	Details
Embedding extraction	
Base models	Qwen2.5-1.5B / Pythia-1.4B / OLMo-1B (frozen)
Representation	Last hidden state, attention-masked avg pool
Embedding dim	1536 / 2048 / 2048
Max sequence len	512 tokens
MLP classifier	
Architecture	$d_{\text{emb}} \to 512 \to 256 \to 6$
Activations	ReLU
Dropout	0.1 (after each hidden layer)
Output	6-class softmax
Training	
Optimizer	Adam (lr $= 10^{-3}$, weight decay $= 10^{-4}$)
Loss	Cross-entropy
Batch size	256
Max epochs	20
Early stopping	Patience = 5 (validation loss)
Cross-validation	5-fold GroupKFold

Table 17: Pairwise AUC with bootstrap 95% confidence intervals for four model-mode configurations (Exp-007, 1,000 bootstrap resamples). The horizontal rule separates depth pairs above and below the detectability threshold $\theta = 0.60$. Permutation p-values test H_0: AUC $= 0.50$. All pairs are significant at $p < 0.01$ except OLMo 1 vs. 2 through 4 vs. 5 (all $p > 0.09$), consistent with $K^*_{\text{empirical}}$=1 for OLMo.

Depth Pair	Qwen-rewrite	Pythia-1.4B	OLMo-1B	Qwen-cont
0 vs. 1	.996 [.995, .997]	.993 [.992, .994]	.986 [.984, .988]	.997 [.996, .998]
1 vs. 2	.696 [.686, .704]	.654 [.645, .663]	.508 [.496, .519]	.611 [.602, .621]
2 vs. 3	.589 [.580, .599]	.551 [.542, .563]	.497 [.486, .509]	.561 [.551, .571]
3 vs. 4	.544 [.533, .555]	.515 [.504, .526]	.502 [.491, .513]	.541 [.531, .551]
4 vs. 5	.534 [.524, .545]	.514 [.502, .525]	.494 [.483, .506]	.514 [.504, .526]

Each cell: AUC [95% CI]. CI from 1,000 bootstrap resamples; p-values from 500-resample permutation test. All pairs $p < 0.01$ except OLMo 1v2–4v5 ($p > 0.09$).

Table 18: Spearman rank correlation (ρ) between binary detector confidence scores and true generational depth, and 6-class depth classification accuracy (Exp-012). Three metrics: log-PPL from the reference model (DivEye), log-PPL from the observer model, and PPL ratio (Binoculars; observer: Pythia-1.4B, performer: OLMo-1B). Random baseline for 6-class accuracy: 0.167. While statistically significant, correlations are driven by the 0v1 boundary; confidence scores do not monotonically track depth beyond generation 1. Qwen-rewrite exhibits reversed sign ($\rho > 0$), indicating that generation mode alters the direction of binary confidence's relationship with depth.

	Qwen-rw	Pythia	OLMo	Qwen-cont
Spearman ρ with true depth				
log-PPL (ref)	-0.123	-0.297	-0.378	-0.389
log-PPL (observer)	$+0.082$	-0.453	-0.461	-0.162
PPL ratio	$+0.119$	-0.392	-0.407	-0.058
Classification				
6-class accuracy	0.199	0.130	0.127	0.183
Random baseline: 0.167	$K^*_{\text{empirical}}{}^{\text{binary}} = 1$		$K^*_{\text{empirical}}{}^{\text{OBD}} = 2$	

Table 19: Pairwise AUC with bootstrap 95% CIs across depth pairs for Pythia-1.4B under four decoding configurations (Exp-020; 10,000 bootstrap resamples). Complements Table 8 (Qwen-rewrite) with a second model family. †Greedy decoding triggers repetition collapse in Pythia-1.4B (4-gram repeat rate: d_0=0.01 $\rightarrow$ d_1=0.72), inflating 0v1 and 1v2 AUC via lexical degeneracy rather than depth-discriminative features.

Depth Pair	Greedy†	Nuc p=0.9, T=1.0	Nuc p=0.95, T=0.7	Nuc p=0.95, T=1.2
0 vs. 1	.999 [.999, .999]	.995 [.994, .996]	.996 [.995, .997]	.994 [.993, .994]
1 vs. 2	.884 [.878, .890]	.675 [.667, .684]	.728 [.721, .736]	.618 [.608, .628]
2 vs. 3	.531 [.520, .541]	.559 [.549, .570]	.598 [.589, .606]‡	.531 [.520, .541]
3 vs. 4	.522 [.511, .532]	.524 [.513, .534]	.563 [.553, .572]	.520 [.509, .531]
4 vs. 5	.514 [.504, .525]	.501 [.490, .512]	.527 [.518, .537]	.510 [.499, .521]
$K^*_{\text{empirical}}$	2	2	2	2

‡ CI upper bound (0.606) exceeds θ=0.60; point estimate (.598) below threshold.

Table 20: Pairwise AUC across depth pairs for three segment lengths (Exp-021; Qwen2.5-1.5B base, continuation mode). $K^*_{\text{empirical}}$=2 is confirmed for 128 and 256 tokens; 512 tokens yields point estimate $K^*_{\text{empirical}}$=2 but the 1v2 bootstrap CI lower bound falls marginally below θ.

Depth Pair	128 tokens	256 tokens	512 tokens
0 vs. 1	.9791	.9914	.9917
1 vs. 2	.6219	.6282	.6073†
2 vs. 3	.5291	.5556	.5721
3 vs. 4	.5349	.5298	.5300
4 vs. 5	.5012	.5137	.5176
$K^*_{\text{empirical}}$	2	2	2†

†CI [0.5979, 0.6166]; lower bound <θ=0.60.

Table 21: Pairwise AUC across depth pairs for three text domains (Qwen2.5-1.5B-Instruct, continuation mode). Tests generalization of $K^*_{\text{empirical}}$=2 beyond C4 web text to structured and technical domains.

Depth Pair	**C4 (web)**	**Wikipedia**	**arXiv**
0 vs. 1	.9971	.9961	.9948
1 vs. 2	.6112	.6193	.8581
2 vs. 3	.5609	.5498	.6436 $[.628, .658]^*$
3 vs. 4	.5409	.5231	.5440
4 vs. 5	.5141	.5141	.5299
$K^*_{\text{empirical}}$	2	2	3

*Bootstrap 95% CI (n=10,000); permutation $p < 0.001$. CI lower bound exceeds θ.

Table 22: Scorer sensitivity: $K^*_{\text{empirical}}$ under self-scoring vs. cross-scoring. Self-scorer uses the generator's own model family for feature extraction; cross-scorer uses alternative models (Pythia-1.4B, OLMo-1B). $K^*_{\text{empirical}}$ agreement across scorer conditions indicates the boundary is governed by the generation process. Cross-scoring reduces $K^*_{\text{empirical}}$ only for the most marginal self-scorer condition (Qwen-continuation).

		Scorer			
Generator	**Mode**	**Self**	**Pythia**	**OLMo**	**Agreement**
Qwen-1.5B-Instruct	Rewrite	2	2	2	✓
Qwen-1.5B-Instruct	Continuation	2	1	1	×
Qwen-1.5B (base)	Continuation	2	2	2	✓
Pythia-1.4B	Continuation	2	2	2	✓
OLMo-1B	Continuation	1	—	—	—
Qwen-7B	Continuation	2	—	—	—
Mistral-7B	Continuation	1	—	—	—

D Extended Discussion

D.1 Theoretical Implications of $K^*_{\text{empirical}} = 2$

The DPI guarantees a finite K^*_{theory}; for general web text, $K^*_{\text{empirical}} \leq 2$ across all 10 configurations (1B–8B, 6 families), while structured domains (arXiv) yield $K^*_{\text{empirical}}$=3, aligning with model collapse convergence (Shumailov et al., 2024; Dohmatob et al., 2024; Alemohammad et al., 2024). Decoding strategy modestly affects signal strength (1v2 AUC $\in [0.6866, 0.7039]$) but does not shift the boundary.

D.2 Cross-Model Transfer Implications

Cross-model classification retains 80–100% accuracy with scorer-invariant feature rankings ($\rho > 0.90$), enabling single-classifier deployment across generators. Absolute discriminability at the $K^*_{\text{empirical}}$ boundary is scorer-sensitive: alternative scorers reduce Qwen-continuation $K^*_{\text{empirical}}$ from 2 to 1, while rewrite-mode $K^*_{\text{empirical}} = 2$ is preserved. Model family similarity modulates transfer (Pythia↔OLMo: 90–92% vs. Qwen-cont↔OLMo: 80–85%).

Table 23: Cross-scorer validation on Qwen-generated text. Alternative scorers (Pythia-1.4B, OLMo-1B) preserve feature ranking ($\rho > 0.90$) and rewrite-mode $K^*_{\text{empirical}} = 2$, but reduce continuation-mode $K^*_{\text{empirical}}$ from 2 to 1.

Scorer	Gen. Data	K^*	AUC(0v1)	AUC(1v2)	Spearman ρ
Qwen-1.5B (self)	Qwen-RW	2	—	0.6955	—
Pythia-1.4B	Qwen-RW	2	0.9923	0.7012	0.906
OLMo-1B	Qwen-RW	2	0.9918	0.6947	0.920
Qwen-1.5B (self)	Qwen-cont	2	—	0.6112	—
Pythia-1.4B	Qwen-cont	1†	0.9900	0.5923	0.942
OLMo-1B	Qwen-cont	1†	0.9890	0.5858	0.902
Qwen-1.5B	Qwen-7B-cont	2	0.9644	0.6178	—

Table 24: Three-way ANOVA (Generator × Scorer × Depth-Pair) on per-fold pairwise AUC values. DV = per-fold AUC; N=300 (12 cells × 5 folds × 5 depth-pairs). Scorer is statistically significant but explains modest variance (η_p^2=.166) relative to Depth-Pair (η_p^2=.997), supporting near-scorer-independence of depth discrimination.

Source	df	F	p	η_p^2
Generator	2	851.91	<.001	.866
Scorer	3	17.48	<.001	.166
Depth-Pair	4	22737.70	<.001	.997
Gen×Scorer	6	7.50	<.001	.146
Gen×Depth	8	913.41	<.001	.965
Scorer×Depth	12	2.25	.010	.093
Residual	264			

D.3 Feature Importance Across Decoding Strategies

Feature importance ordering varies across decoding strategies: greedy and T=0.7 share highly correlated rankings ($\rho = 0.914$), while greedy vs. T=1.2 yields lower correlation ($\rho = 0.53$, non-significant after Bonferroni). The depth boundary is robust but the optimal feature subset is strategy-dependent.

D.4 Scale and Neural Probe

Cross-scale validation confirms $K^*_{\text{empirical}} = 2$ from 1.5B through 8B: Qwen-7B (1v2 AUC = 0.6168 $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC = 0.6123, 0v1 = 0.9225) both exceed θ, with progressively lower 0v1 signal consistent with larger models generating more human-like text. Scaling to 13B+ remains open.

The neural probe yields model-dependent $K^*_{\text{empirical}}{}^{\text{neural}}$: 3 for Qwen (self-familiarity effect from instruction-tuned representations) but 2 for Pythia; OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC = 0.5054), consistent with the 15D LightGBM boundary. The 15D model-free features yield $K^*_{\text{empirical}} = 2$ for Pythia and Qwen, while OLMo yields $K^*_{\text{empirical}} = 1$ under the standardized pipeline.

D.5 LLM Zero-Shot Baseline

GPT-5.5 achieves 33.2% accuracy (MAE = 1.10) and Claude Opus 4.6 33.1% (MAE = 1.15) on a balanced 500-sample test set, comparable to OBD-LightGBM's 38.0% (95% CI: $[0.340, 0.422]$; McNemar $p = 0.081, 0.069$; Table 10). Per-depth recall confirms convergent failure at depths ≥ 2 (Section 6). When 6-class probabilities are marginalized to 5 classes ($P(d{\geq}4)$), accuracy rises to 44.4% but concentrates in depth-4 (recall: 22%→80%) while depths 2–3 collapse to 2–8%; we report raw 6-class results.

D.6 Practical Considerations

All filtering strategies degrade general capabilities (MMLU: 0.6018 → 0.5865–0.5941; HellaSwag: 0.6821 → 0.6638–0.6701); our 100% synthetic setting is an upper bound; in mixed corpora, degradation scales with contamination fraction (Shumailov et al., 2024; Dohmatob et al., 2024). K^*-aware binary filtering reduces MMLU degradation by ∼50% (−1.53 pp → −0.77 pp), requires no per-model calibration, and is combinable with distribution-matching methods (DSIR, DoReMi).

D.7 Extended Limitations

Prompt-chain versus retrain-chain. Retrain-chain validation (Pythia-1.4B LoRA, 5 generations) yields $K^*_{\text{empirical}} \geq 4$ (1v2 AUC $= 0.9484$, 2v3 $= 0.8888$, 3v4 $= 0.7982$), far exceeding prompt-chain $K^*_{\text{empirical}}=2$. The elevated signal arises from cumulative model-adaptation artifacts rather than text-intrinsic degradation; prompt-chain $K^*_{\text{empirical}}$ is the operationally relevant boundary for data curation.

Threshold sensitivity. At $\theta = 0.55$, $K^*_{\text{empirical}} = 3$ for Qwen-rewrite/continuation; at $\theta = 0.70$, $K^*_{\text{empirical}} = 1$ across all configurations (Figure 4). Qwen-continuation's 1v2 CI lower bound exceeds θ by only 0.2 pp; all ten boundary comparisons yield permutation $p < 0.001$.

Structural confound: token length. In continuation mode the generation prompt is discarded, so all depths ≥ 1 consist entirely of machine-generated tokens and the 1v2 boundary compares texts differing only in conditioning context (human-authored vs. machine-generated prompt), not structural composition. Depth 0 texts average $\sim$191 subword tokens (C4 truncation), while depth 1+ texts average 193–256 tokens depending on model (Table 12). This asymmetry is directionally opposite to a length-inflation confound: shorter depth-0 texts provide less feature-extraction context, making 0v1 discrimination harder, not easier. Within depths 1–5, mean token counts are stable (within-model range ≤ 4 tokens for all configurations except Qwen-continuation at 6 tokens), so the pairwise comparisons underlying $K^*_{\text{empirical}}=2$ are unaffected by length. EOS truncation rates are similarly depth-stable within each model: Qwen-rewrite 7–11%, Pythia 34–36%, OLMo <1%, Qwen-continuation 17–20%, Qwen-7B 45–47%. Models with the highest truncation (Pythia, Qwen-7B) still yield $K^*_{\text{empirical}}=2$ with the same step-function AUC decay pattern as low-truncation models (Table 2), confirming that EOS truncation does not drive the boundary.

Graduated filtering. All α values (0.1–0.9) underperform binary-ours on MMLU (α=0.1: 0.5873; α=0.9: 0.5892 vs. 0.5941); at α=0.9, HellaSwag matches exactly (0.6701). Binary-ours significantly outperforms all alternatives on MMLU ($p_{\text{corr}} \leq 0.021$); Graduated$_{0.9}$ is indistinguishable on HellaSwag ($p = 1.00$) but worse on MMLU ($p_{\text{corr}} = 0.011$).

D.8 Dose-Response: Contamination Ratio Effect

The main filtering experiments (Section 5) use 100% synthetic data to maximize effect size. To characterize behavior at lower contamination ratios, we mix synthetic depths 1–5 with clean C4 at 10%, 30%, and 50% contamination (Qwen-1.5B continuation, 5 seeds each; Table 25).

Table 25: MMLU accuracy by contamination ratio (fraction of synthetic data in training set) and filtering strategy. Base model (no fine-tuning): 0.6018. Mean ± std over 5 seeds.

Strategy	10%	30%	50%
No-filter	$0.6009 \pm .003$	$0.5978 \pm .004$	$0.5932 \pm .004$
Binary-ours	$0.6011 \pm .003$	$0.5988 \pm .003$	$0.5956 \pm .004$
Graduated	$0.6008 \pm .003$	$0.5982 \pm .004$	$0.5944 \pm .004$

At 10% contamination, no-filter MMLU (0.6009) remains within noise of the base model (0.6018, $\Delta = -0.09$ pp), and all three strategies are indistinguishable ($\Delta_{\max} = 0.03$ pp between strategies, $p > 0.8$). At 30%, differences emerge directionally ($\Delta = 0.10$ pp, binary vs. no-filter) but remain non-significant ($p > 0.3$). At 50%, binary-ours shows a 0.24 pp advantage over no-filter, approaching but not reaching significance ($p = 0.12$). The strategy ordering (binary > graduated > no-filter) is consistent with the 100% setting, but effect sizes scale linearly with contamination fraction. This demonstrates that K^*-aware curation provides practical benefit primarily when contamination is severe enough to cause measurable quality degradation.

D.9 Seed Robustness

Because the random seed governs both corpus sampling and generation quality (not merely CV splits), absolute AUC values are seed-dependent. We re-run three configurations with seed = 123 (primary: seed = 42; Table 26). $K^*_{\text{empirical}}$ is stable in 2/3 cases: Pythia-1.4B on Wikipedia yields $K^*_{\text{empirical}}=1$ under both seeds (1v2 $\Delta = -0.8$ pp), and Qwen-1.5B on C4 yields $K^*_{\text{empirical}}=2$ (1v2 $\Delta = +9.0$ pp, both seeds above θ). Qwen-1.5B on Wikipedia shifts from $K^*_{\text{empirical}}=2$ to $K^*_{\text{empirical}}=3$ under seed = 123, but the 2v3 AUC (0.6067) is marginal: one of five folds falls below θ (fold 5 $= 0.595$), so this does not constitute strong evidence for $K^*_{\text{empirical}}=3$. The ±9 pp absolute AUC variation across seeds strengthens the methodological case for threshold-based $K^*_{\text{empirical}}$ over raw

AUC comparisons: the qualitative boundary is robust even when point estimates shift.

Table 26: Seed robustness: $K^*_{\text{empirical}}$ under two random seeds. 1v2 Δ: change in 1v2 AUC from seed 42 to seed 123. $K^*_{\text{empirical}}$ is stable in 2/3 configurations; the Qwen-Wiki shift to $K^*_{\text{empirical}}$=3 is marginal (2v3 fold variance crosses θ).

Model	Domain	$K^*_{\text{empirical}}$		1v2 Δ	Stable?
		seed=42	seed=123		
Pythia-1.4B	Wikipedia	1	1	-0.8 pp	✓
Qwen-1.5B	C4	2	2	$+9.0$ pp	✓
Qwen-1.5B	Wikipedia	2	3†	$+8.7$ pp	×

†2v3 AUC $= 0.6067$; fold $5 = 0.595 < \theta$. Marginal boundary.

D.10 Bonferroni-Corrected Confidence Intervals

To control the family-wise error rate across all 10 model configurations, we apply Bonferroni correction: $\alpha_{\text{corr}} = 0.05/10 = 0.005$, yielding 99.5% confidence intervals ($z = 2.807$). Table 27 reports the corrected intervals for each configuration's $K^*_{\text{empirical}}$ boundary pair. Eight of ten configurations retain CI lower bounds above θ=0.60; two marginal $K^*_{\text{empirical}}$=2 configurations (Qwen-cont, LLaMA-8B) see their lower bounds dip below θ under the stricter criterion (0.5976 and 0.5988, respectively). These near-misses reflect narrow effect sizes (AUC $- \theta < 0.013$) rather than classification failure: both retain point estimates above θ and uncorrected 95% CI lower bounds above θ.

Table 27: Bonferroni-corrected 99.5% confidence intervals for $K^*_{\text{empirical}}$ boundary pairs across all 10 configurations. $\alpha_{\text{corr}} = 0.005$; CI computed as AUC $\pm\, 2.807 \times \hat{\sigma}_{\text{boot}}$. "Pass θ": CI lower bound > 0.60.

Configuration	Boundary	AUC	$\hat{\sigma}_{\text{boot}}$	99.5% CI	Pass θ
$K^*_{empirical} = 2$ configurations					
Qwen-rw (Instruct)	1v2	.6955	.0046	[.683, .708]	✓
Pythia-1.4B	1v2	.6539	.0046	[.641, .667]	✓
Qwen-base	1v2	.6260	.0047	[.613, .639]	✓
Qwen-7B‡	1v2	.6168	.0046	[.604, .630]	✓
LLaMA-8B	1v2	.6123	.0048	[.599, .626]	×
Qwen-cont (Instruct)	1v2	.6112	.0049	[.598, .625]	×
$K^*_{empirical} = 1$ configurations					
OLMo-1B	0v1	.9860	.0009	[.983, .989]	✓
Mistral-7B	0v1	.9842	.0009	[.982, .987]	✓
Pythia-6.9B§	0v1	.9599	.0015†	[.956, .964]	✓
Gemma-2-2B	0v1	.9530	.0018	[.948, .958]	✓

‡Generator: Qwen2.5-7B; scorer: Qwen2.5-1.5B.
§Nucleus sampling (p=0.95, T=1.0).
†Estimated from analogous configurations (no separate bootstrap JSON).

D.11 Cross-Sample Robustness: 5K vs. 10K

The preceding analyses fix the sample pool at N=5,000 passages per depth. To assess between-sample variability, we re-draw an independent N=10,000 sample from C4 using a different random seed and repeat the full OBD pipeline. If $K^*_{\text{empirical}}$ and the boundary AUC are driven by the underlying distributional shift rather than by idiosyncrasies of a particular sample, both quantities should remain stable across draws.

Table 28 reports the results. For Pythia-1.4B (continuation, C4), the 5K and 10K samples agree on $K^*_{\text{empirical}}$=2: the 0v1 AUC is nearly identical (0.9700 vs. 0.9712), while the boundary 1v2 AUC decreases from 0.6539 to 0.6175 ($\Delta = -0.036$). This shift is consistent with natural sampling variability across C4 subsets and does not alter the qualitative depth boundary. The 2v3 pair remains near chance in both draws (0.54, 0.54), confirming that depths ≥ 2 are indistinguishable regardless of sample. Results for Qwen-1.5B and LLaMA-3.1-8B (continuation, C4) are pending; once available, they will allow a more comprehensive assessment of cross-sample stability across model scales.

Table 28: Cross-sample robustness: pairwise AUC and $K^*_{\text{empirical}}$ under 5K and 10K C4 samples drawn with different random seeds. All configurations use continuation mode. $K^*_{\text{empirical}}$ is determined by the θ=0.60 threshold on 1v2 AUC.

Model	N	AUC(0v1)	AUC(1v2)	AUC(2v3)	$K^*_{\text{empirical}}$
Pythia-1.4B cont	5K	0.9700	0.6539	0.5400	2
Pythia-1.4B cont	10K	0.9712	0.6175	0.5364	2
Qwen-1.5B cont	5K		(see Table 27)		2
Qwen-1.5B cont	10K		*pending*		
LLaMA-8B cont	5K		(see Table 27)		2
LLaMA-8B cont	10K		*pending*		